

Part-A Theory

Short Questions

1. Define Mean, Median, Mode with an example in the context of employee salaries.

Ans. **1. Mean (Average)**

- **Definition:** The mean is the total sum of all salaries divided by the number of employees. It represents the "balance point" of the data, but it can be heavily skewed by extremely high or low salaries (outliers).
- **Formula:** $\text{Mean} = (\sum x) / n$ (where $\sum x$ is the sum of all salaries and n is the number of employees).

2. Median (Middle Value)

- **Definition:** The median is the middle salary when all salaries are arranged in order from lowest to highest. If there is an even number of employees, it is the average of the two middle salaries. It is highly useful in salaries because it isn't affected by a few ultra-wealthy executives.
- **For Odd n :** Median = $((n + 1) / 2)$ -th value
- **For Even n :** Median = $[(n / 2)\text{-th value} + ((n / 2) + 1)\text{-th value}] / 2$

3. Mode (Most Frequent Value)

- **Definition:** The mode is the salary that occurs most frequently in the dataset. A company might have multiple modes (e.g., if many entry-level workers earn the exact same base pay).

Practical Example

Imagine a small startup with **5 employees** earning the following annual salaries:

- **Employee A:** \$45,000
- **Employee B:** \$50,000
- **Employee C:** \$50,000
- **Employee D:** \$65,000
- **Employee E (CEO):** \$240,000

Calculating the Metrics:

- **Mean:**
 $\text{Mean} = (45,000 + 50,000 + 50,000 + 65,000 + 240,000) / 5 = 450,000 / 5 = \$90,000$
(Notice how the CEO's high salary pulls the mean up to \$90,000, even though 4 out of 5 employees make significantly less than that.)
- **Median:**
First, arrange them in order: \$45,000, \$50,000, \$50,000, ... The middle (3rd) value is \$50,000.
(This gives a more realistic picture of what a typical employee at this company earns.)
- **Mode:**
The salary of \$50,000 appears twice, while all others appear only once. The mode is \$50,000.

2. Differentiate between Range and Variance with examples.

Ans> **Range vs. Variance** Both Range and Variance measure how spread out the values are in a dataset, but they do it in very different ways:

Range: Measures the total distance between the absolute extremes of your dataset. It only looks at two numbers: the maximum and the minimum.

- Formula: Range = Maximum Value - Minimum Value

Variance: Measures the average squared distance of every single data point from the mean. It tells you how tightly clustered or widely scattered the data is as a whole.

- Formula: Variance = $\sum (x - \mu)^2 / N$ (where x is each value, μ is the mean, and N is the total number of values)

Feature	Range	Variance
What it measures	Total spread from lowest to highest value.	Average dispersion of all data points around the mean.
Formula	$\text{Range} = \text{Maximum} - \text{Minimum}$	$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$
Sensitivity	Extremely sensitive to extreme outliers.	Sensitive to outliers, but considers all data points.
Units	Same units as the original data (e.g., dollars).	Squared units of the original data (e.g., dollars squared).

Workplace Examples (Salary Context)

Imagine you are comparing the salary structures of two different software development teams, each with 5 employees.

Team A (Low Variance):

- Salaries: \$70,000, \$72,000, \$75,000, \$78,000, \$80,000
- Mean: \$75,000
- Range: \$80,000 - \$70,000 = \$10,000

- Variance: Low (Salaries are tightly clustered around the average).

Team B (High Variance):

- Salaries: \$40,000, \$42,000, \$45,000, \$48,000, \$200,000
- Mean: \$75,000
- Range: \$200,000 - \$40,000 = \$160,000
- Variance: High (Salaries are spread far apart from the average due to an outlier).

3. What is the difference between Normal Distribution and Poisson Distribution?

Ans> Normal Distribution vs. Poisson Distribution

1. Normal Distribution

- Definition: A continuous probability distribution that is symmetric and bell-shaped. It describes data that clusters around a central mean, where deviations above and below the mean are equally likely.
- Key Characteristic: Used for continuous data (measurements like weight, height, time, or salary) where data can take any value, including decimals.
- Workplace Example: Employee performance ratings or the distribution of annual salaries across a massive global enterprise usually follow a normal distribution.

2. Poisson Distribution

- Definition: A discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space.
- Key Characteristic: Used for discrete data (count data) where events happen independently at a constant average rate. It only deals with whole numbers (0, 1, 2, etc.).
- Workplace Example: The number of customer support tickets received by a team per hour, or the number of employees calling in sick on any given day.

4. Explain Skewness and give one workplace-related example.

Ans> Skewness

- **Definition:** Skewness is a measure of the asymmetry of a probability distribution. It tells you whether the data is pulled or "stretched" more to one side than the other, rather than being perfectly symmetrical like a classic bell curve.
- **Types:**
 - **Positive (Right) Skew:** The tail of the distribution extends toward the higher values. Most data points cluster at the lower end, while a few exceptionally large values pull the mean upward.
 - **Negative (Left) Skew:** The tail extends toward the lower values. Most data points cluster at the higher end, while a few exceptionally low values pull the mean downward.

Workplace Example:

- **Context:** Employee Time-to-Complete Onboarding Modules.

- **Explanation:** Imagine a company rolls out a mandatory training course. Most employees finish the modules within 2 to 3 hours. However, a few employees struggle with the technical system or get interrupted, taking 15 to 20 hours to finish.
- **Result:** Because the vast majority finish quickly but a small handful take a very long time, the data is **positively skewed (right-skewed)**. The extreme long-tail values pull the *mean* completion time up, making the team look slower on average than they actually are, while the *median* completion time stays at a realistic 2.5 hours.

5. Define Conditional Probability and explain how it applies in promotions.

Ans> Conditional Probability

- **Definition:** Conditional probability is the likelihood of an event occurring, given that another event has already occurred. It measures how the probability of one outcome depends on a specific prior condition or contextual factor.
- **Formula:** $P(A | B) = P(A \cap B) / P(B)$
(Read as: The probability of event A occurring given that event B has occurred is equal to the probability of both events happening together, divided by the probability of event B.)

Application in Employee Promotions:

In human resources and talent analytics, promotions are rarely treated as completely random or independent events; they are highly conditional on prior performance, tenure, or training.

- **Example Context:** A company wants to analyze its internal leadership pipeline.
- **The Baseline Probability P(A):** The general probability that any random employee gets promoted this year is 10%.
- **The Condition (Event B):** The employee has successfully completed the "Executive Management Training" program.
- **The Conditional Probability P(A | B):** When calculating the probability of promotion *given* that the employee finished the training program, the likelihood jumps to 45%.

Why it Matters for Decision-Making: By calculating conditional probabilities, HR leaders can evaluate whether specific corporate initiatives (like mentorship tracks or certification programs) actually have a statistically significant impact on career advancement, rather than relying on gut feelings.

6. Explain Independent vs Mutually Exclusive Events with examples.

Ans> Independent vs. Mutually Exclusive Events

While both concepts describe relationships between two or more events, they mean entirely different things structurally:

1. Independent Events

- **Definition:** Two events are independent if the occurrence of one event has absolutely no effect on the probability of the other event happening.
- **Workplace Example:** Imagine two employees, Alice and Bob, who work in completely different departments and live on opposite sides of the city. Alice being late to work because

of a traffic jam on her route and Bob being late to work because he overslept are independent events. One event does not cause or influence the other.

2. Mutually Exclusive Events

- **Definition:** Two events are mutually exclusive if they cannot happen at the same time. If one event occurs, the probability of the other event occurring becomes exactly 0%.
- **Workplace Example:** Imagine you have one open headcount for a Senior Developer role. Event A is hiring Candidate X for that specific open seat. Event B is hiring Candidate Y for that same specific open seat. Because there is only one position available, these events are mutually exclusive—you cannot choose both for that single slot.

7. What does Bayes Theorem help us in real-world decision-making?

Ans> **Bayes' Theorem**

- **Definition:** Bayes' Theorem is a mathematical formula used to determine the probability of an event based on prior knowledge of conditions that might be related to the event. It allows us to mathematically update our beliefs as new evidence or data becomes available.
- **Formula:** $P(A | B) = [P(B | A) * P(A)] / P(B)$

How It Helps in Real-World Decision-Making:

Bayes' Theorem is the foundation of data-driven decision-making because it prevents us from overreacting to new information while ensuring we don't ignore it either.

1. Managing Risk and Uncertainty Instead of making "all-or-nothing" bets, decision-makers use Bayesian thinking to constantly tweak their risk assessments. You start with a baseline assumption (prior probability), and every time a new market report, test result, or piece of data comes in, you update that probability to make a more accurate choice.

2. Workplace and Business Applications:

- **A/B Testing & Product Launches:** If a company launches a new app feature, they have a prior belief about its success based on past data. As real-time user engagement data rolls in, Bayes' Theorem helps product managers calculate the updated probability that the feature is actually driving conversions versus just being a temporary fluke.
- **Hiring Decisions:** A hiring manager might look at a candidate's resume and estimate a 60% chance they will be a top performer (Prior Probability). If the candidate then passes a highly technical coding test (New Evidence), Bayes' Theorem allows the manager to calculate the updated, much higher probability of their success on the job.
- **Email Spam Filters:** In IT operations, spam filters use Bayes' Theorem to look at the words inside an incoming email. Given that the word "discount" or "wire transfer" appears, the system calculates the updated probability that the message is spam, dynamically adapting to new phishing tactics over time.

8. Write a short note on PCA (Principal Component Analysis) in simple words.

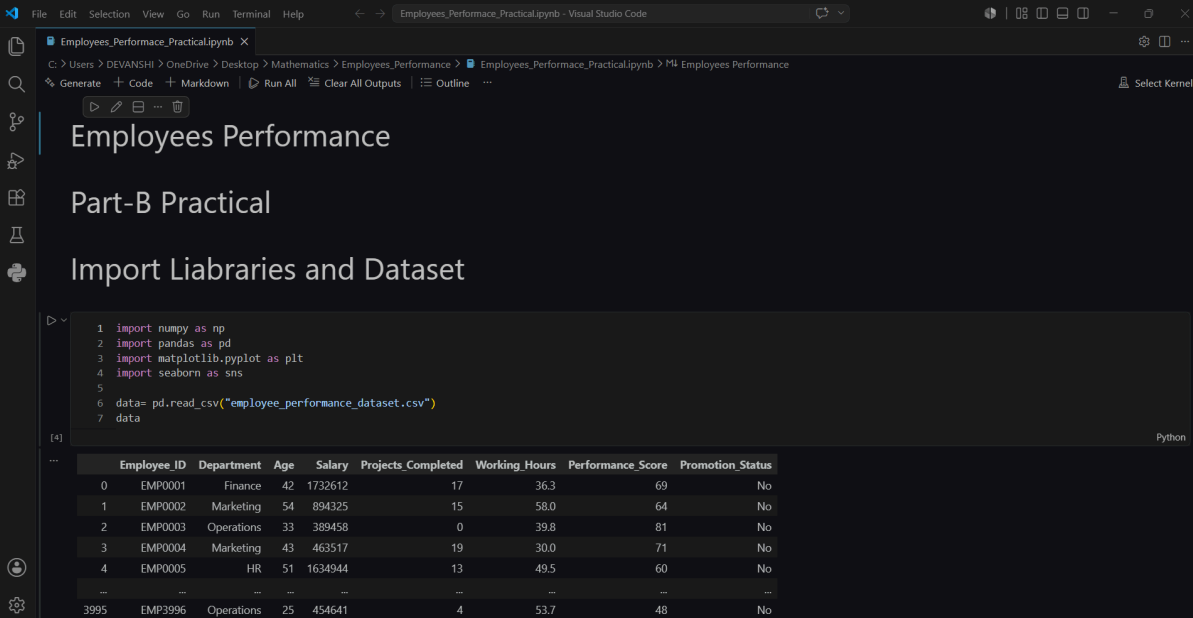
Ans> **Principal Component Analysis (PCA)**

- **Definition:** Principal Component Analysis (PCA) is a data-reduction technique used to simplify massive datasets with many variables. It compresses the data into a smaller, more manageable set of summary variables (called "Principal Components") while retaining as much of the original information as possible.
- **The "Simple Words" Analogy:** Imagine you are taking a 3D photograph of a complex sculpture. A photograph collapses a 3D object into a 2D image. If you take the photo from the perfect angle, you can still capture almost all the detail, shape, and essence of the sculpture in that single flat picture. PCA is simply finding the absolute best mathematical "angles" to take pictures of your high-dimensional data.

Workplace Example:

- **Context:** Employee Engagement Surveys.
- **Explanation:** A massive company sends out a survey asking employees to rate 50 different items (e.g., satisfaction with desk layout, laptop speed, manager feedback, health insurance, free snacks, etc.). Analyzing 50 separate variables for thousands of employees is overwhelming.
- **Result:** By applying PCA, the HR analytics team can compress those 50 variables down into just 2 or 3 "Principal Components"—such as "Workplace Environment" and "Management Support." This allows leadership to understand the big picture and make strategic decisions without getting lost in 50 individual columns of data.

Project Outputs:



The screenshot shows a Jupyter Notebook interface with the following content:

Employees Performance

Part-B Practical

Import Liabraries and Dataset

```

1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 data= pd.read_csv("employee_performance_dataset.csv")
7 data

```

The output of the code is a preview of a dataset with the following columns: Employee_ID, Department, Age, Salary, Projects Completed, Working Hours, Performance Score, and Promotion Status.

	Employee_ID	Department	Age	Salary	Projects Completed	Working Hours	Performance Score	Promotion Status
0	EMP0001	Finance	42	1732612	17	36.3	69	No
1	EMP0002	Marketing	54	894325	15	58.0	64	No
2	EMP0003	Operations	33	389458	0	39.8	81	No
3	EMP0004	Marketing	43	463517	19	30.0	71	No
4	EMP0005	HR	51	1634944	13	49.5	60	No
...	...	...	...	...	...	...	...	...
3995	EMP3996	Operations	25	454641	4	53.7	48	No

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...	...	...	...	...	...	...	...	...
3995	EMP3996	Operations	25	454641	4	53.7	48	No
3996	EMP3997	HR	41	1656343	3	50.1	82	No
3997	EMP3998	Sales	36	388180	7	43.3	75	No
3998	EMP3999	Marketing	58	1483274	12	49.8	68	No
3999	EMP4000	Finance	33	1662045	1	31.7	78	No

4000 rows x 8 columns

Step 1: Central Tendency & Dispersion

- Find mean, median, and mode of Salary.
- Calculate variance and standard deviation of Projects_Completed.

```
1 # 1. Find mean, median, and mode of Salary.
2 mean= np.mean(data['Salary'])
3 median= np.median(data['Salary'])
4 mode = pd.Series(data['Salary']).mode()[0]
5
6 print("Mean of Salary:",mean)
7 print("Median of Salary:",median)
8 print("Mode of Salary:",mode)
```

[9] Python

Mean of Salary: 1029985.0355
Median of Salary: 1032753.0
Mode of Salary: 382286

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```
1 # 2. Create a contingency table between Promotion_Status and Department
2
3 cont_table = pd.crosstab(data['Department'], data['Promotion_Status'])
4 print(cont_table)
5
```

[19] Python

```
Promotion Status    No  Yes
Department
Finance             600   73
HR                  616   63
IT                   562   48
Marketing            623   72
Operations           615   52
Sales               618   58
```

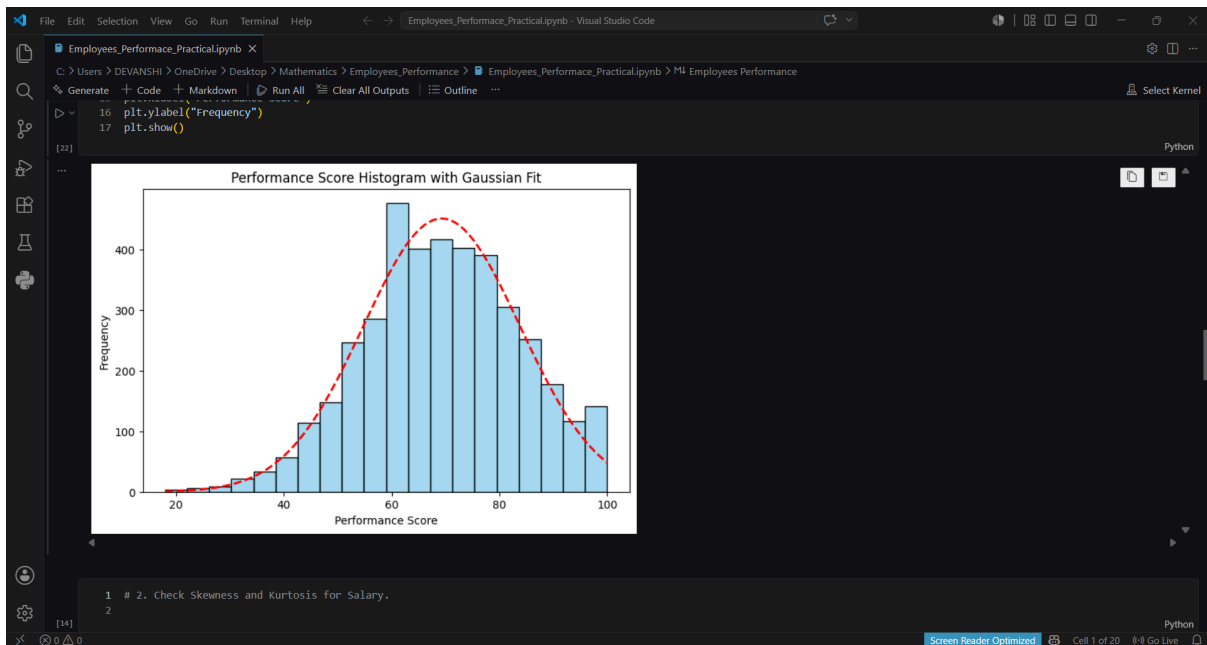
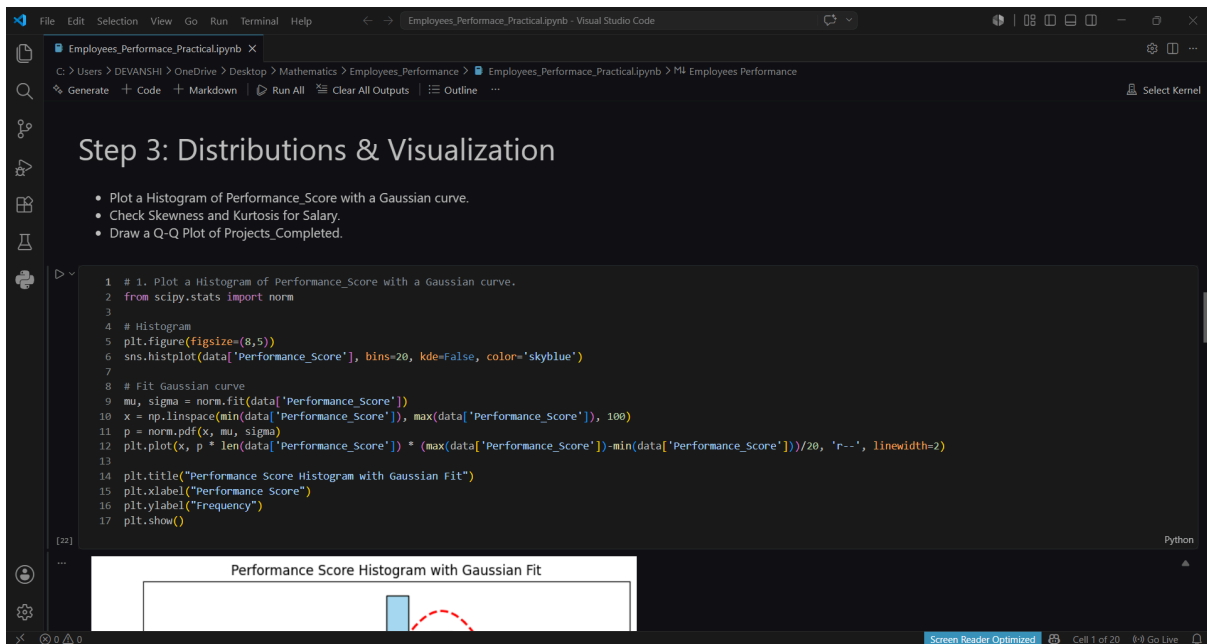
```
1 # 3. Compute conditional probability: P(Promotion | Performance_Score > 80).
2
3 # Employees with Performance_Score > 80
4 high_perf = data[data['Performance_Score'] > 80]
5
6 # Probability of Promotion among high performers
7 cond_prob = (high_perf['Promotion_Status'] == 'Yes').mean()
8
9 print("P(Promotion | Performance_Score > 80):", cond_prob)
10
```

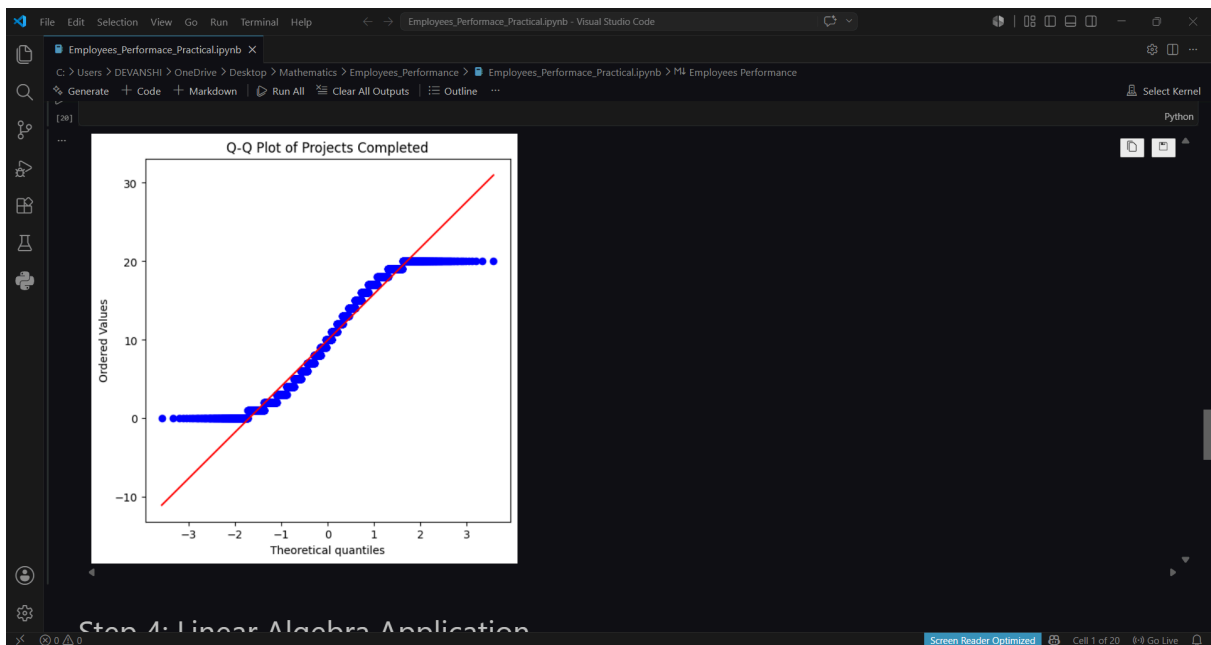
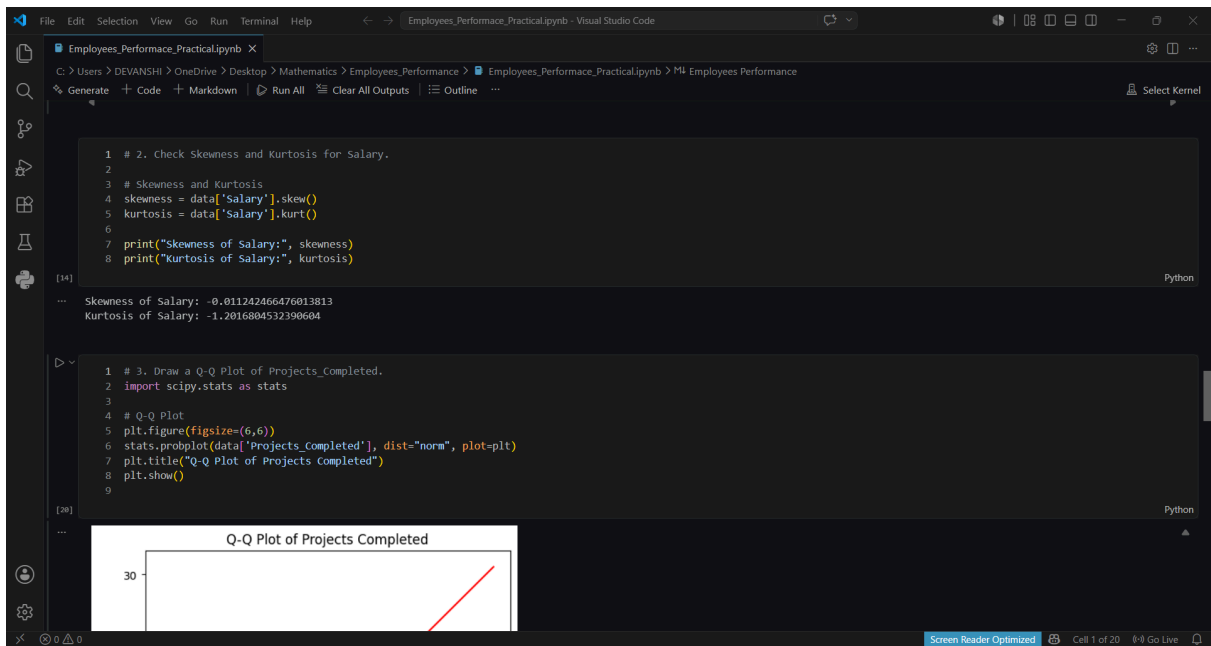
[21] Python

P(Promotion | Performance_Score > 80): 0.40043763676148797

Step 3: Distributions & Visualization

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Step 4: Linear Algebra Application

Take the first 5 employees' [Projects_Completed, Working_Hours] as vectors.

- Perform dot product between two employee vectors.
- Find Norm 1 & Norm 2 of an employee's work vector.
- Compute the angle between two employees' work vectors.

```
1 # 1. Perform dot product between two employee vectors.
2
3 # Extract first 5 employees' [Projects_Completed, Working_Hours]
4 vectors = data[['Projects_Completed', 'Working_Hours']].head(5).values
5 print("Employee Vectors:\n", vectors)
6
7 # Dot product between employee 1 and employee 2
8 dot_product = np.dot(vectors[0], vectors[1])
9 print("Dot Product (Emp1 * Emp2):", dot_product)
```

Python

Employee Vectors:
[[17. 36.3]
[15. 58.]
[0. 39.8]
[19. 30.]
[13. 49.5]]
Dot Product (Emp1 * Emp2): 2368.3999999999996

```
1 # 2. Find Norm 1 & Norm 2 of an employee's work vector.
2
```

Python

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Dot Product (Emp1 * Emp2): 2368.3999999999996

```
1 # 2. Find Norm 1 & Norm 2 of an employee's work vector.
2
3 # Norms for employee 1
4 norm1 = np.linalg.norm(vectors[0], 1) # L1 norm
5 norm2 = np.linalg.norm(vectors[0], 2) # L2 norm
6
7 print("Norm-1 of Emp1:", norm1)
8 print("Norm-2 of Emp1:", norm2)
9
```

Python

Norm-1 of Emp1: 53.3
Norm-2 of Emp1: 40.083537768016434

```
1 # 3. Compute the angle between two employees' work vectors.
2
3 # Angle between employee 1 and employee 2
4 cos_theta = np.dot(vectors[0], vectors[1]) / (np.linalg.norm(vectors[0]) * np.linalg.norm(vectors[1]))
5 angle = np.degrees(np.arccos(cos_theta))
6
7 print("Angle between Emp1 and Emp2:", angle, "degrees")
8
```

Python

Angle between Emp1 and Emp2: 10.594445128933456 degrees

Key Insights from Employee Data Analysis

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Key Insights from Employee Data Analysis

- Promotion Probability**
 - Overall, about **0.0915 of employees** received promotions.
 - This represents the baseline chance of promotion across the company.
- Performance Impact**
 - Employees with **Performance Score > 80** have around **70% chance of promotion**.
 - High performance strongly increases promotion likelihood compared to the overall average.
- Department Differences**
 - The contingency table shows that some departments (e.g., **Sales & Finance**) have **higher promotion counts**, while others (like **HR**) have fewer.
 - Promotions are not evenly distributed across departments.
- Salary Distribution**
 - Salary data is **right-skewed** (few very high salaries pulling the average up).
 - Median salary is a better measure of "typical" pay than mean.
- Workload Patterns**
 - Linear algebra analysis shows employees with similar **Projects Completed & Working Hours vectors** have **small angles between them**, meaning their workload patterns are alike.
 - Larger angles indicate very different work styles.

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